

MIPS Online

Time: 20-25 minutes

Section B

```
int splitScore(int *A, int low, int high, int bias) {
    if (high < low) return 0;
    if (high == low) {
        if (A[low] ≤ bias) return A[low];
        else return -A[low];
    }

    int mid = (low + high) / 2;
    int left = splitScore(A, low, mid, bias);
    int right = splitScore(A, mid + 1, high, bias);
    return left + right + A[mid];
}
```

Section C

```
int arrayScore(int A[], int low, int high){
    if (low > high) return 0;
    int sum = 0;
    for(int i = low; i ≤ high; i++) {
        sum += A[i]*(i - low + 1);
    }
    int left = arrayScore(A, low-1, high);
    int right = arrayScore(A, low, high+1);
    return left - right + sum + A[low] + A[high];
}
```

Section A

```
int arrayScore(int A[], int low, int high, int factor){
    if (low > high) return 0;
    int sum = 0;
    for(int i = low; i ≤ high; i++) {
        if(A[i] > high){
            sum+= A[i] * (i - low +1);
        }
        else if(A[i] ≥ low){
            sum += A[i] * i
        }
        else{
            sum -= A[i] * factor
        }
    }
    int mid = (low + high)/2
    int left = arrayScore(A,mid,high+1, factor-1)
    return left+sum+A[mid];
}
```